Katlego kgone

SD3 Exercise 3 Fundamentals

(1) SELECT product_name, price,

CASE

WHEN price < 100 THEN 'Budget'

WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'

ELSE 'Expensive'

END AS Price-category

FROM products;

Product_name	Price	Price_category
Laptop	1200.00	Expensive
Phone	850.00	Mid-range
keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

(2) SELECT customer_name,

amount,

CASE

WHEN amount >= 1000 THEN 'High value'

WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium value'

ELSE 'Low value'

END AS order_value_category

FROM orders

Customer-name	amount	order value - category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.56	Low value
Ethan	1200.00	High value

(3)
```
SELECT emp_name, department, salary,
CASE
    WHEN department = 'IT' AND salary > 80 000 THEN 'Senior IT'
    WHEN department = 'HR' AND salary > 55 000 THEN 'Experienced HR'
ELSE 'Staff'
END AS Position_level
FROM employees;
```

emp_name	department	salary	Position_level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	staff
Lucy	Finance	95 000	staff
Tom	HR	55 000	staff

(4)
```
SELECT student_name, score,
CASE
    WHEN score < 60 THEN 'F'
    WHEN score BETWEEN 60 and 67 THEN 'D'
    WHEN score BETWEEN 70 AND 79 THEN 'C'
    WHEN score BETWEEN 80 AND 89 THEN 'B'
  ELSE 'A'
  END AS grade
FROM students;
```

Student_name	Score	grade
Ann	92	A
Ben	76	C
Cora	59	F
David	83	B
Ella	68	D

(5)
```
SELECT delivery_id, delivery_time_minutes,
CASE
    WHEN delivery_time_minutes <= 30 THEN 'Fast'
    WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On time'
  ELSE 'Late'
END AS performance
FROM deliveries;
```

delivery_id	delivery_time_minutes	Performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

(6)

```
SELECT issue_type, priority,
CASE
  WHEN Priority = 3 THEN 'High'
  WHEN Priority = 2 THEN 'Medium'
  Else 'Low'
END AS Priority_label
FROM tickets;
```

Issue type	Priority	Priority_label
Login issue	1	Low
Server down	3	High
Slow System	2	Medium
Email error	2	Medium
Password reset	1	Low

(7)

```
SELECT student_id,
       (days_present * 100.0 / total_days) AS attendance_percentage,
CASE
    WHEN (days_present * 100.0 / total_days) >= 90 THEN 'Excellent'
    WHEN attendance_percentage BETWEEN 75 and 89 THEN 'Good'
ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

(8)

```
SELECT product_id,
       stock_qty,
CASE
    WHEN stock_qty = 0 THEN 'Out of Stock'
    WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
ELSE 'In Stock'
END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

(9)

```
SELECT subject, enrolled_students,
CASE
WHEN enrolled_students >=25 THEN 'Large'
WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
ELSE 'Small'
END AS class_size_category
FROM classes;
```

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

(10)

```
SELECT payment_id,
       Payment_method,
       amount,
CASE
    WHEN Payment_method = 'cash' AND amount >= 200
        THEN 'Eligible for Discount'
    ELSE 'Not Eligible'
END AS discounts_eligibility
FROM payments;
```

Payment_id	Payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount